Educational Phlogiston

Why a Formula Is Never Just a Formula

On symbolic notation, referential binding, and the teaching of physics equations

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Cognitive load is a genuine constraint, but invoked wherever interpretation fails it becomes an invisible explanatory substance: educational phlogiston. The five equivalent forms of Newton's second law show why a sharper diagnosis is available: identical syntax persists across entirely different physical situations, and the notation carries no visible marker for the mapping between symbol and world that the student must supply unaided.

Consider the five familiar faces of one law:

$$F = ma, \qquad F_{\text{net}} = ma, \qquad \sum F = ma,$$
$$\sum_i \mathbf{F}_i = m\mathbf{a}, \qquad \sum_i \mathbf{F}_i = \frac{d\mathbf{p}}{dt}.$$

Each form is, in one sense, a strict improvement: it specifies that the relevant force is the net force, that several interactions may contribute, that force and acceleration are vectors, that ma is a constant-mass restriction of a more general momentum law. Yet the symbol F occupies the same syntactic slot in all five, whether it denotes a single spring force or the resultant of gravity, friction and a normal reaction acting on a block. The notation does not change shape to tell the reader which physical situation is in play; that mapping has to be supplied from outside the equation, every time, by someone who already knows how.

This is the actual difficulty, and it is not principally a matter of how many symbols must be held in working memory. School algebra trains students to treat a symbol's meaning as either given outright or irrelevant to the manipulation required; $F = ma$ looks identical to that training, so students default to reading it the way they read algebra, as a self-interpreting template, rather than as a slot that must be re-instantiated, by physical reasoning, on every new occasion.

The reversal error

Clement's classic study made this visible in miniature. Asked to translate "there are six times as many students as professors" into algebra, many otherwise capable students wrote $6S = P$ rather than $S = 6P$, matching symbol order to word order rather than treating the coefficient as a multiplier relating two measured quantities [1]. The letters were being read as labels for nouns, not as variables denoting numbers. $F = ma$ invites exactly this misreading: a student can parse it as the sentence "force is mass times acceleration" without ever asking which force, produced by what interaction, acting on which system, along which axis. The algebra is satisfied by the sentence. The physics is not.

Symbolic forms, and what chunking actually is

Sherin's account of expert reading describes *symbolic forms*: recurring notational patterns fused, for the expert, with a conceptual schema, so that ma is read directly as inertia coupled to a rate of change, and $\sum_i \mathbf{F}_i$ as distinct interactions accumulating into a resultant [3]. This fusion is not a smaller quantity of the same load; it is a pre-built instantiation procedure, assembled through exactly the recovery work that Torigoe calls *unpacking*, in which a learner reconstructs the physical commitments a compressed equation has hidden [5]. The novice experiences the equation as sparse because that procedure has not yet been built; the expert experiences it as compressed because it has. Redish's observation that physics uses mathematics as a distinct disciplinary dialect, in which form and physical meaning are continuously blended, is really this same point from a linguistic angle [4].

Take a block sliding down a rough incline. The left-hand side of $\sum_i \mathbf{F}_i = m\mathbf{a}$ is a placeholder for three physically unlike contributions:

$$\mathbf{F}_{\text{net}} = \mathbf{F}_{\text{gravity}} + \mathbf{F}_{\text{normal}} + \mathbf{F}_{\text{friction}}.$$

Gravity depends on position in a field; the normal reaction depends on a contact constraint; friction depends on the normal reaction's magnitude and the direction of relative sliding. None of this heterogeneity is visible in the summation sign, which treats three different kinds of physical fact as interchangeable terms in an addition. A formula sheet supplies the equation; it does not supply the decision that friction here acts up the slope, opposing the block's slide, or that the normal reaction is not simply equal and opposite to weight once the surface is tilted.

What the diagram supplies that the algebra hides

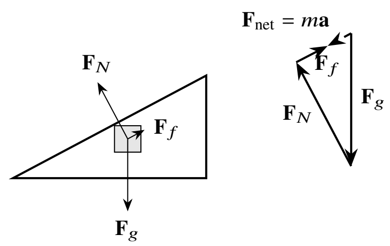

Figure 1: Left: forces on the block, drawn from a common origin. Right: the same vectors placed tail to tip. The path does not close; the dashed gap between start and end *is* the net force, and its direction fixes the direction of **a**.

The tail-to-tip construction does something the algebra structurally cannot: it makes non-closure visible. If the three vectors returned to their starting point, the block would be in equilibrium, and the diagram would show a closed polygon. Because they do not return, the gap is not a residue or an error; it is the net force, drawn as a specific vector with a specific direction, in the same picture as the forces that produced it.

Chandler and Sweller's split-attention research explains why this helps: it is costly to search and mentally integrate a symbolic sum and a physical situation across separated representations [2]. The diagram earns its keep by encoding, spatially, the one thing the summation sign conceals, namely which quantities are being combined and what their combination looks like. A word-for-symbol gloss (as in a GCSE formula booklet) placed beside $F = ma$ cannot do this, because it duplicates the notation's emptiness rather than filling it.

"The gap in the diagram is the net force."

The diagnosis, restated

Cognitive load is real, but naming it explains little on its own. The more useful question is narrower: has this student built the procedure that instantiates $\sum_i \mathbf{F}_i = m\mathbf{a}$ for a specific system, in a specific frame, with specific interactions identified and a specific axis chosen, or are they still reading the equation the way school algebra taught them to, as a syntactic template that interprets itself? The five equivalent forms look identical regardless of which is true. A tail-to-tip diagram, by contrast, fails visibly the moment a force has been misidentified or mis-signed, which is exactly why it belongs in front of the student before the algebra does, not beside it afterwards. Without that distinction, cognitive load risks becoming educational phlogiston still: an invisible explanatory substance inserted wherever meaning failed to ignite.

References

[1] J. Clement, "Algebra word problem solutions: Thought processes underlying a common misconception," *Journal for Research in Mathematics Education*, vol. 13, no. 1, pp. 16–30, 1982. doi: 10.2307/748434.

[2] P. Chandler and J. Sweller, "The split-attention effect as a factor in the design of instruction," *British Journal of Educational Psychology*, vol. 62, no. 2, pp. 233–246, 1992. doi: 10.1111/j.2044-8279.1992.tb01017.x.

[3] B. L. Sherin, "How students understand physics equations," *Cognition and Instruction*, vol. 19, no. 4, pp. 479–541, 2001. doi: 10.1207/S1532690XCI1904_3.

[4] E. F. Redish, "Problem solving and the use of math in physics courses," in *Proceedings of the Conference World View on Physics Education in 2005: Focusing on Change*, Delhi, India, pp. 1–10, 2005. Available: arXiv:physics/0608268.

[5] E. T. Torigoe, "Unpacking symbolic equations in introductory physics," in *Physics Education Research Conference 2015*, College Park, MD: American Association of Physics Teachers, pp. 327–330, 2015. doi: 10.1119/perc.2015.pr.078.

Physics equations are compressed models. Fluency consists in knowing what has been compressed, and in demanding a representation that shows it.